Symmetric & Asymmetric Encryption with ~~Example~~ Explanation of AES, Diffie-Hellman and Digital Signature.

Chat application end to end encryption

https.

Encryption:

Cipher

- It's a process of converting readable data into text which is difficult to understand.

- It make use of "Cryptographic key". Consider it as a mathematical value which encryption Method uses to convert the readable text into unreadable format.

For example

Key: 2

HIMANSHU → Encryption Method → FRQPHSW FRGLQJ

Key: 2

FRQFHSW FRGLQJ → Decryption Method → HIMANSHU

Concept && Coding

Encryption is ~~basical~~ a two type:

1. Symmetric Encryption
2. Asymmetric Encryption

1: Symmetric Encryption :→

- Same key is used for both Encryption & Decryption.
- Encryption & Decryption Algorithms :

(Data Encryption Standard)

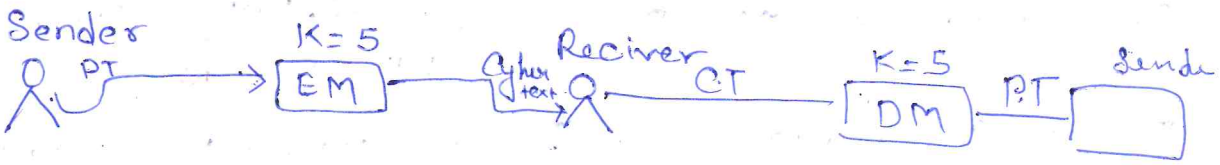
△ DES (NOT RECOMMENDED) (In 2005, It will be cracked because the key length is 64 bits)

1 Fixed key length : 56 bits
2 Process data is 64 bits
3 Perform 16 rounds in its encryption process.

✡ △ AES (Advance Encryption Standard) ✡ ✡ Imp (Biggy the bits more security but Computation more it will be slower also)

1. Support key length : 128, 192 or 256 bits
2. Process data in 128 bits.
3. Encryption rounds depends on key size :
   (a) 10 rounds - 128 bits key
   (b) 12 rounds - 192 bits key
   (c) 14 rounds - 256 bits key

Concept && Coding



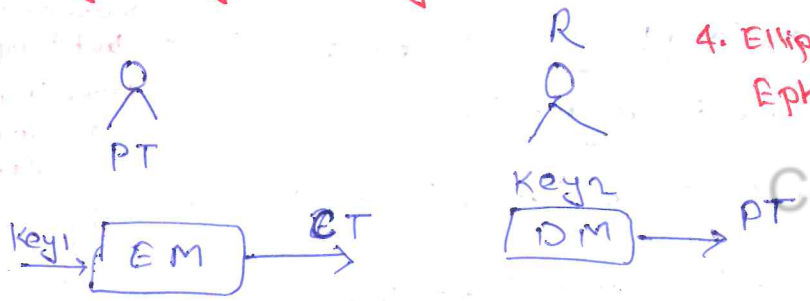
- Advantages :
  (1) Fast and less computation compared to Asymmetric.
  (2) Good for Encrypting Bulk data. (Like chat encryption)
- Disadvantages :
  (1) Key Distribution i.e. how to securely distribute the key between client and server.
  (2) No. of Client increase, means sever has to manage those no. of symmetric keys and its distribution too.

Asymmetric Encryption :-

- Uses 2 different keys i.e Private and Public key.
  (i) Public key is used by Sender to Encrypt.
  (ii) Private key is used by Receiver to Decrypt.

- Encryption & Decryption Algorithms :-
  1:- RSA ✡ 2048 bits key lengths, more secure
  2:- DSA
  3:- Diffie - Hellman ✡✡
  4:- ECDHE

1. (Rivest - Shamir - Adleman) : RSA
2. (Digital Signature Algorithm) : DSA
4. Elliptic Curve Cryptography (ECC)
4. Elliptic Curve Deffie - Hellman Ephemeral. (ECDHE)



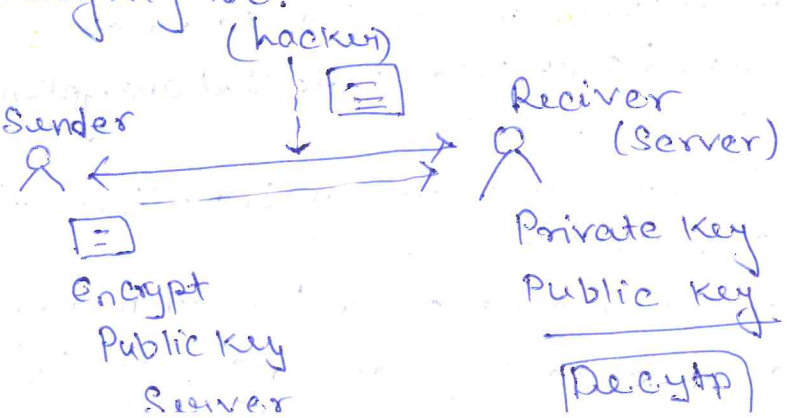
Concept && Coding

- Advantages :-

(1) No Security issues with key Distribution. As each user has secret (private) key, which they do not forward it.

(2) Provide key exchanges protocols like. Diffie - Hellman, ECDHE. ✡

(3) Digital Signature can be created using asymmetric encryption, help to authenticate and provide integrity too.



Disadvantages:

1. Computation intensive compared to Symmetric encryption.
2. Might not suitable for encrypting bulk data as its little slow compare to symmetric.

$(P\ No)^{\wedge\ Key}$ $(5)^{2048}$

AES (Advanced Encryption Standard) (Symmetric Encryption)

- It's a block cipher, means it process the data in the blocks (of 128 bits)

HIMANSHU

1 0 1 0 1 0 1 0 1 0 1 0 1 1

128 bits 128 bits 128 bits

Concept && Coding

Termelogy 1:

Data and keys are arranged like this in 4 x 4 array called "State array";

1 Byte = 8 bits
16 Byte = 16 x 8
= 128 bits

Byte 0	Byte 4	Byte 8	Byte 12
Byte 1	Byte 5	Byte 9	Byte 13
Byte 2	Byte 6	Byte 10	Byte 14
Byte 3	Byte 7	Byte 11	Byte 15

Set of 4 bytes is called "Word"

Key Expansion logic:

Byte 0 ⊗ W_0	Byte 4 ⊗ W_1	Byte 8 ⊗ W_2	Byte 12 ⊗ W_3
Byte 1 ⊗ W_0	Byte 5 ⊗ W_1	Byte 9 ⊗ W_2	Byte 13 ⊗ W_3
Byte 2 ⊗ W_0	Byte 6 ⊗ W_1	Byte 10 ⊗ W_2	Byte 14 ⊗ W_3
Byte 3 ⊗ W_0	Byte 7 ⊗ W_1	Byte 11 ⊗ W_2	Byte 15 ⊗ W_3

W_0	W_1	W_2	W_3

Set of 4 words, called "Round Key".

Key Expansion logic

W_4 = function(W_3) XOR W_0

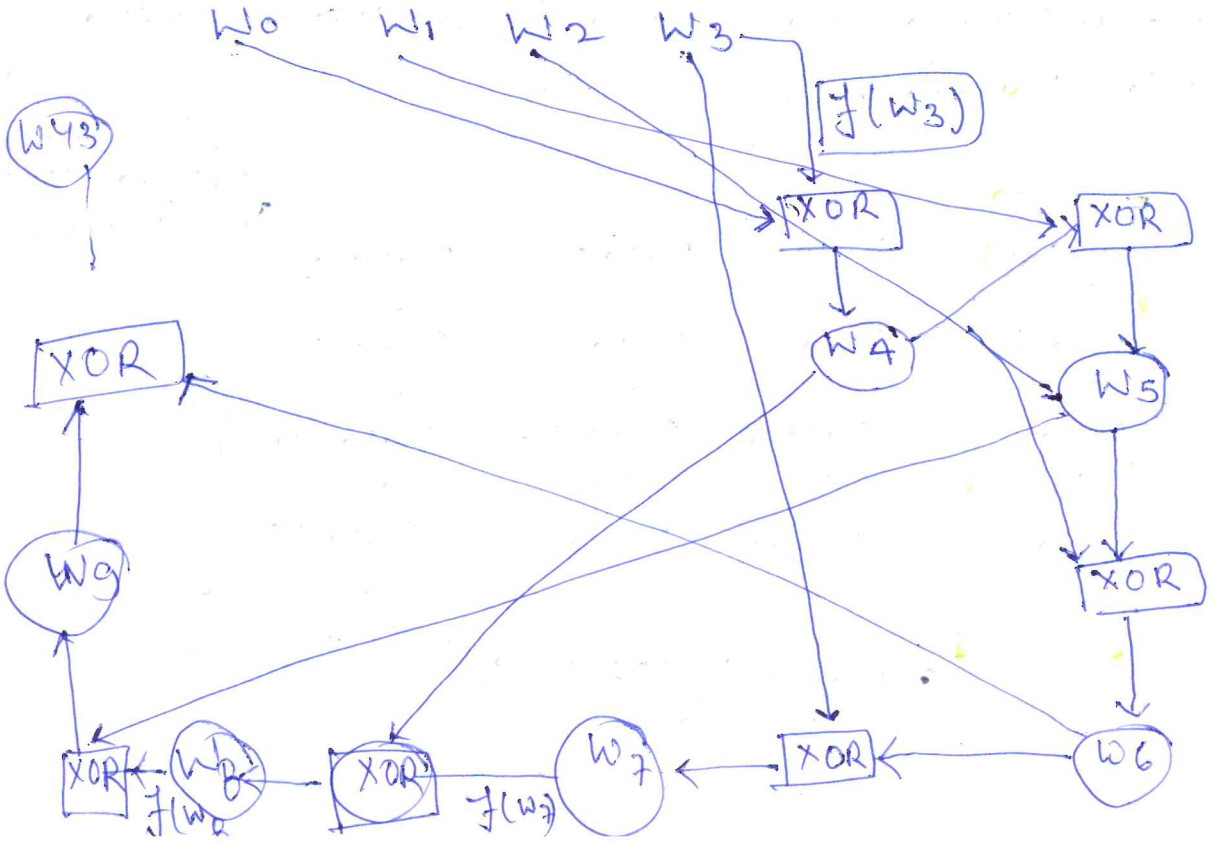
W_5 = W_4 XOR W_1

W_6 = W_5 XOR W_2

W_7 = W_6 XOR W_3

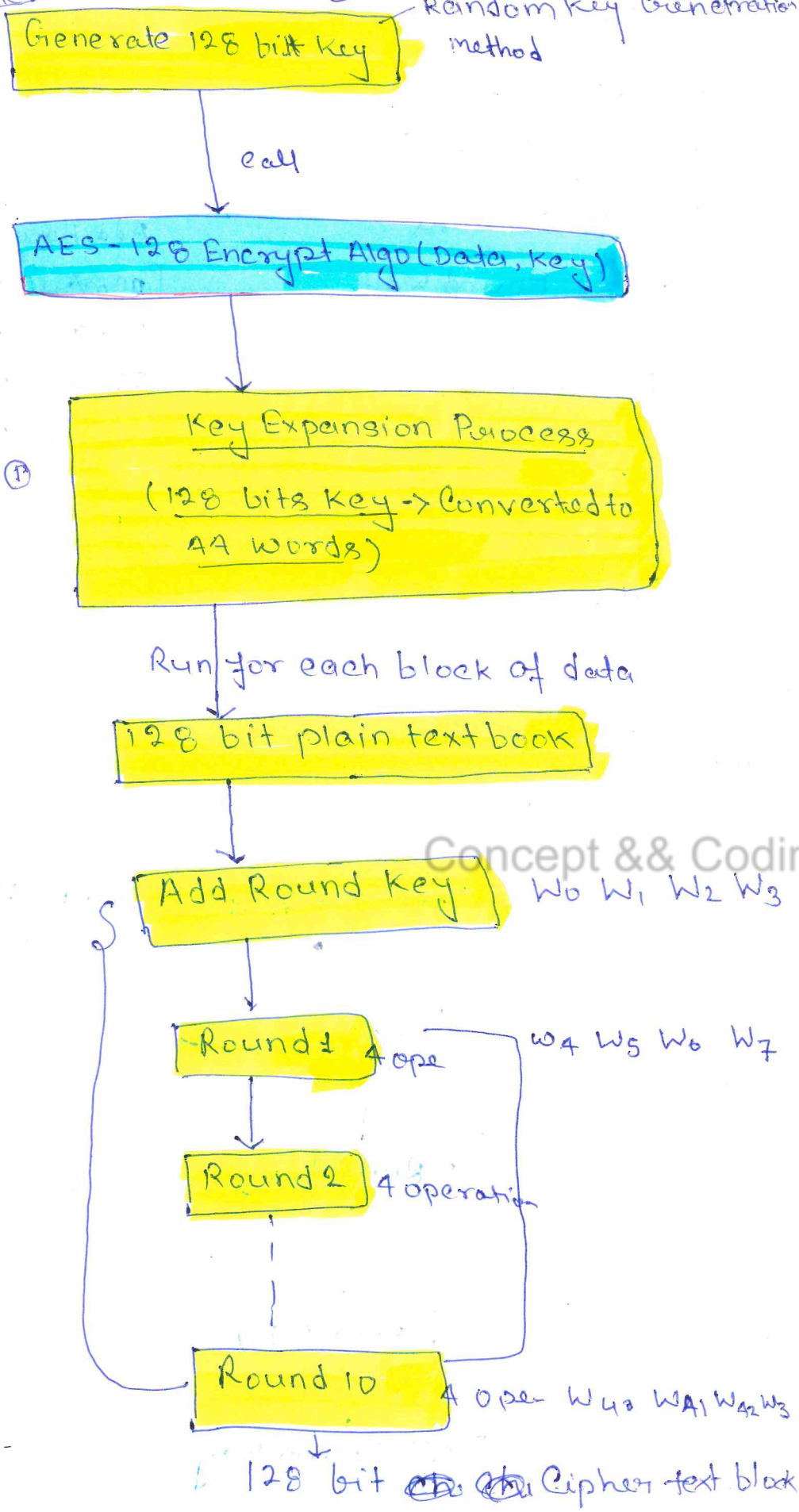
And so on ... till W_{43}

Concept && Coding



Normal to Cipher

(6)



Concept && Coding

Each Round Consists of these Steps :-

1. Substitute Bytes
2. Shift Rows
3. Mix Columns
4. Add Round Key

Add Round Key

$$\begin{bmatrix} DB_0 \otimes W_0 & DB_4 \times W_1 & - & - \\ DB_1 \otimes W_0 & DB_5 \otimes W_1 & - & - \\ DB_2 \times W_0 & DB_6 \otimes W_1 & - & - \\ DB_3 \times W_0 & DB_7 & - & - \end{bmatrix}$$

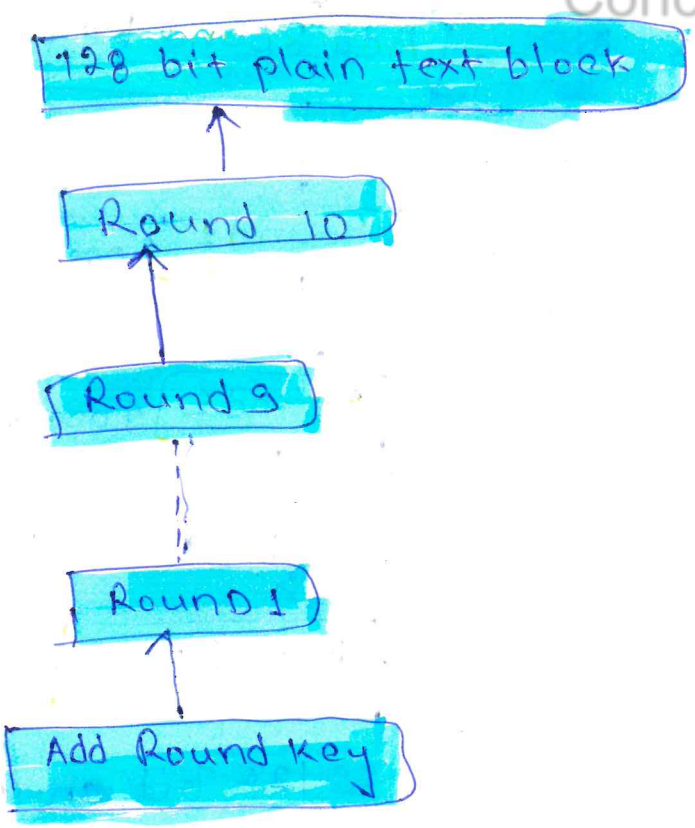
$W_0 \quad W_1 \quad W_2 \quad W_2$

1:- Substitute Bytes

$$\begin{bmatrix} DB_0 & DB_2 & 6 \\ DB_1 & H & 7 \\ DB_2 & S & 8 \end{bmatrix}$$

opposite ~~key~~ cipher to text block

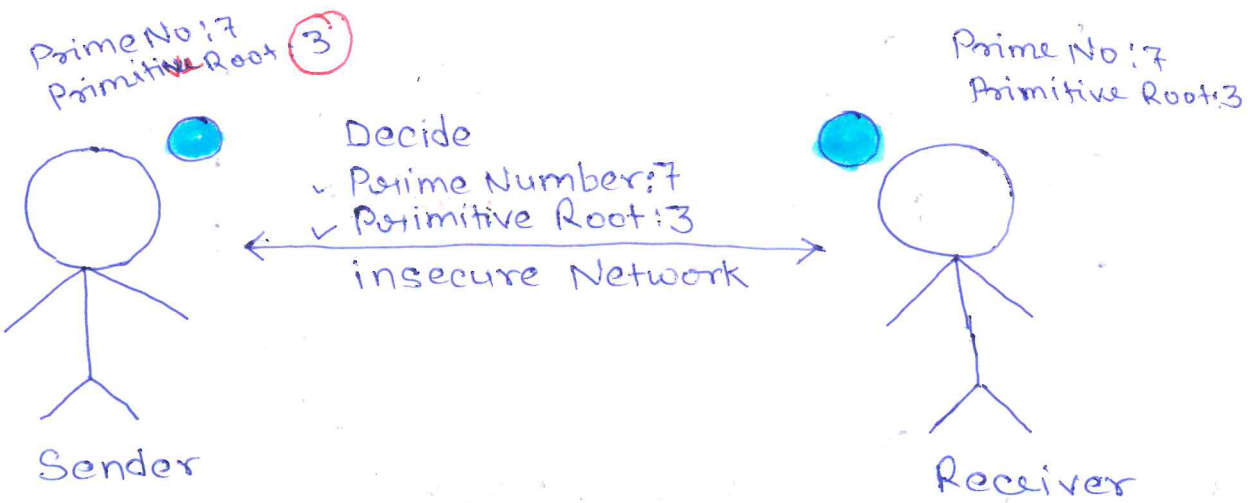
Concept && Coding



128 bit cipher text block

(8) Diffie - Hellman (Asymmetric Encryption) Algorithm

- It's provides a way to share the secret key between **SENDER** and **RECEIVER** over insecure network.



How to find primitive root:

$3^1 \bmod 7 = 3$

$3^2 \bmod 7 = 2$

$3^3 \bmod 7 = 6$

$3^4 \bmod 7 = 4$

$3^5 \bmod 7 = 5$

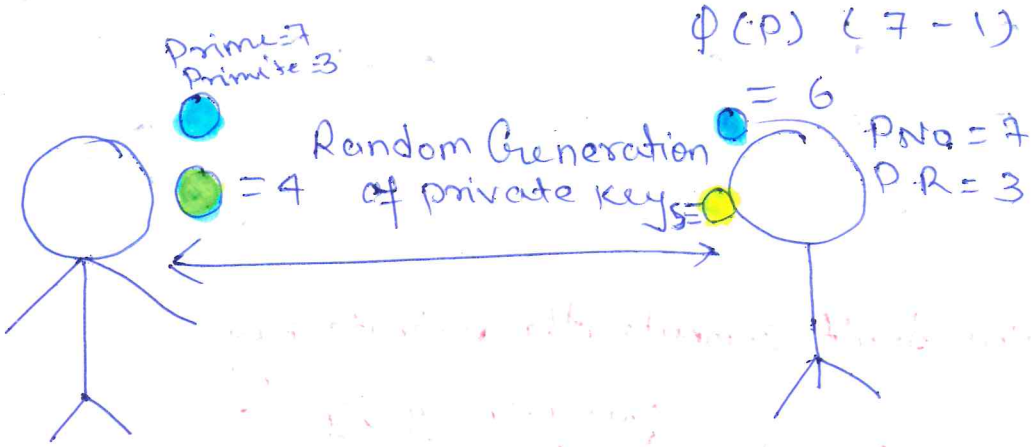
$3^6 \bmod 7 = 1$

$a^i = x \pmod p$

$\phi(p)$ $(7-1)$

$= 6$

Concept && Coding



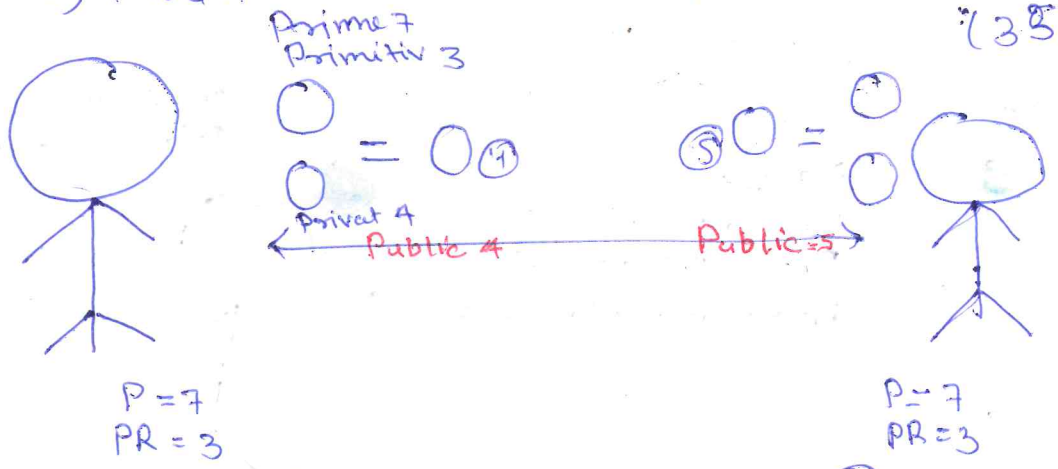
(3) Calculate Public key

(Primitive root ^ Private) Mod prime

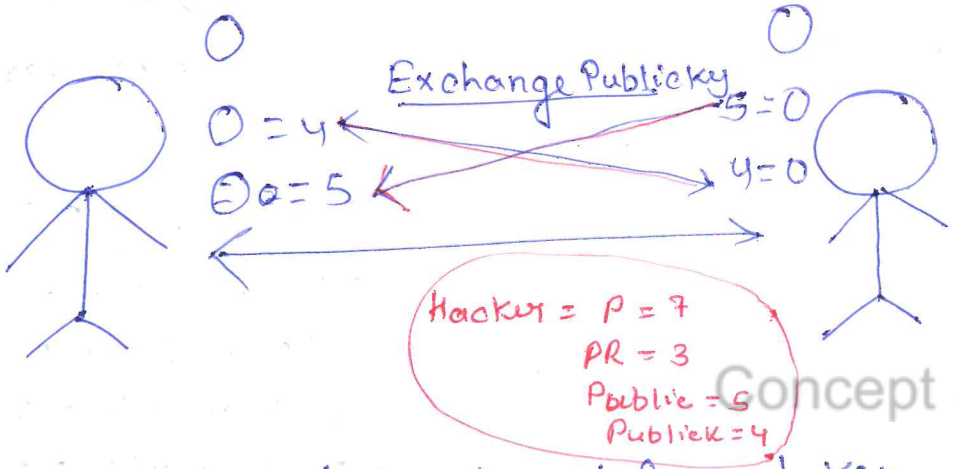
3 4

(3^4) Mod 7 = 4

(3^5) Mod 7 = 5



(4) Exchange Public key



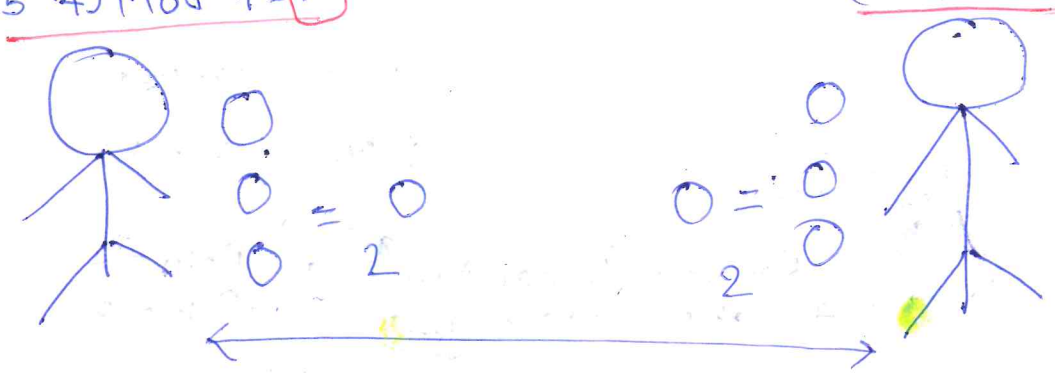
Concept && Coding

(5) Calculate shared secret key

(Others Public ^ private) Mod Prime

(5^4) Mod 7 = 2

(4^5) Mod 7 = 2



Hacker don't compute the private key.

(value) ^ (Private key) Mod N = output

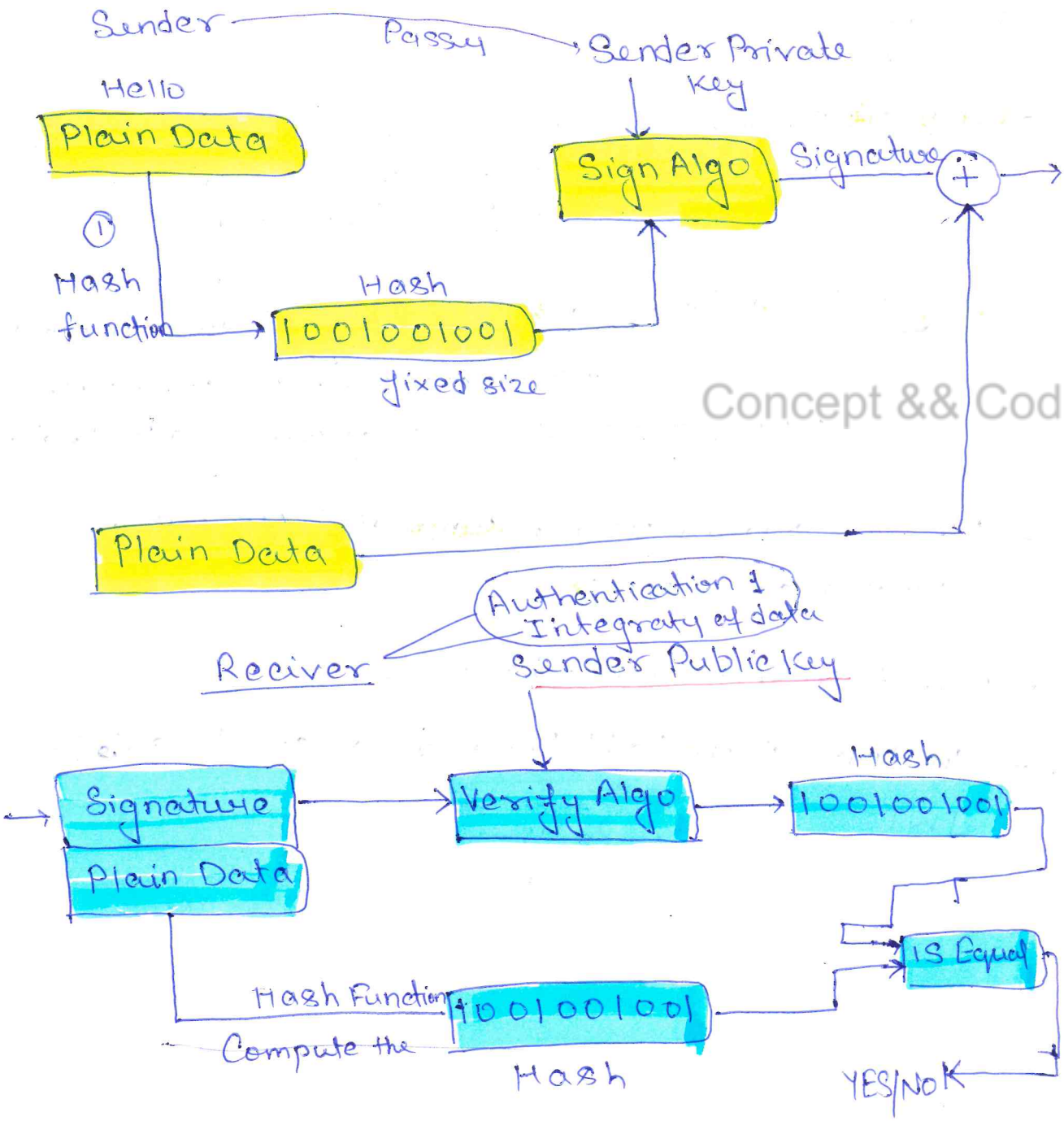
very-very big.

How Digital Signature Works ?

Digital Signature has two propuse

1. Authentication
2. Integraty of data → Data will be not change

Flow of the digital Signature



Concept && Coding